Equivalence testing and Interval hypotheses

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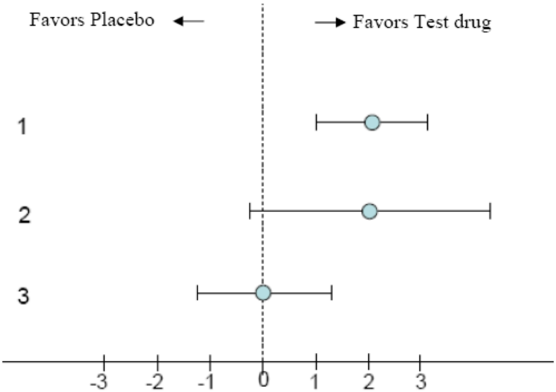
Nil-Null hypothesis significance testing (nil-NHST)

- Most scientific studies are designed to test the prediction that an effect or a difference exists. Does a new intervention work? Is a new intervention superior?
- The goal is to reject the null hypothesis (H_0)
- H_0 is usually set to 0
- If $p < \alpha$ -> we reject H_0
- If $p > \alpha$ -> we fail to reject H_0
- If $p > \alpha$, we cannot claim H_0 is true or there is no effect/difference


Favors Placebo
Favors Test drug
1
2
3
-3
-2
-1
0
1
2
3

Hypotheses about equivalence

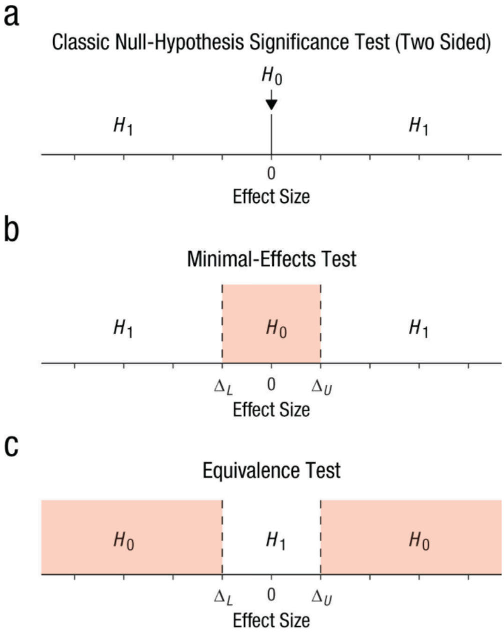
- Researchers sometimes want to show that two interventions are equivalent rather than different
- For example, in pharmacology researchers often want to test whether a cheaper drug works just as well as an existing drug (Senn 2021)

Why can't we claim there is no effect when $p > \alpha$?

https://rpsychologist.com/d3/ci/

- Note that as we increase the sample size, the width of the confidence interval (CI) decreases
- The CI always includes 0 and values around 0 which does not allow to rule out non-zero effects.
- Because it is impossible to rule out an exact effect of 0, we need a range of values (equivalence range) that researchers consider as practically equivalent to 0
- The equivalence range should be specified in advance, and requires careful consideration of the smallest effect size of interest (SESOI)
- The SESOI refers to the smallest effect size that researchers consider practically or theoretically relevant

H_0 in an equivalence test



Equivalence test

- In contrast to nil-NHST, the roles of H_0 and H_1 are reversed
- The goal is to still to reject H_0, however H_0 is not set to 0 but to a range of values (equivalence range)
- The equivalence range is defined by a lower ($\triangle_L$) and upper ($\triangle_U$) bound. Accordingly, two H_0 are formulated:
- H_{01}: μ_1 - $\mu_2 \leq \triangle_L$ (non-inferiority test)
- H_{02}: μ_1 - $\mu_2 \geq \triangle_U$ (non-superiority test)
- Together, these two H_0 represent effects that fall outside the equivalence range
- H_1 corresponds to effects that lie within the range ($\triangle_L$ and $\triangle_U$)

Two-one sided *t-test* (TOST)

- Because equivalence requires rejecting two directional H_0, we use the TOST (two one-sided *t*-tests) procedure (Schuirmann 1987):

$$t_L = \frac{(\mu_1 - \mu_2) - \triangle_L}{\sigma * \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$t_U = \frac{(\mu_1 - \mu_2) - \triangle_U}{\sigma * \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- These two one-sided *t*-tests yield two *p*-values
- If both *p*'s $< \alpha$, we reject effects outside of the predefined equivalence range and conclude equivalence
- The largest p-value is used to report the result of TOST procedure
- TOST is part of the frequentist statistics and requires to control for both type I and type II errors

Potential outcomes

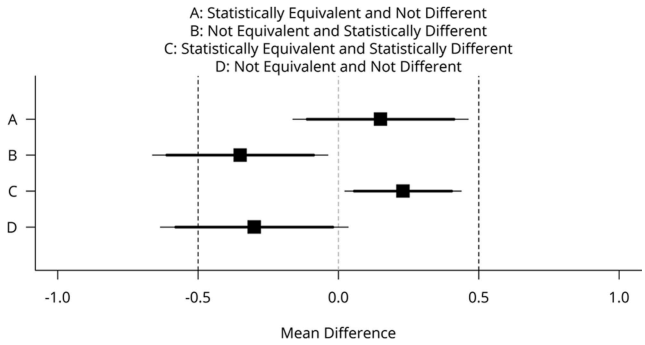


Figure 1. Mean differences (black squares) and 90% confidence intervals (CIs; thick horizontal lines) and 95% CIs (thin horizontal lines) with equivalence bounds $\Delta_L = -.5$ and $\Delta_U = .5$ for four combinations of test results that are statistically equivalent or not and statistically different from zero or not.

Lakens (2017)

Example 1

- Suppose we want to test whether two interventions are equally effective for weight lose
- The equivalence bound ($\triangle$) is set to $\pm$ 2Kg

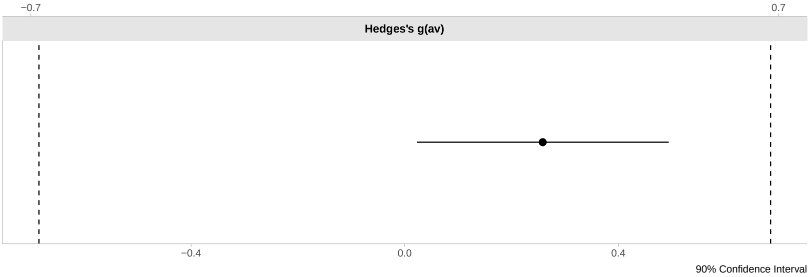
```
set.seed(0009) # for reproducibility

# create two groups
group <- rep(c("INT", "CON"), each = 100)

# Simulate data
weight_data <- data.frame(
  group,
  weight = ifelse(
    group == "INT",
    rnorm(n = 100, mean = 65, sd = 3),
    rnorm(n = 100, mean = 66, sd = 3)
    )
  )
```

	group	weight
1	INT	62.69961
2	INT	62.55062
3	INT	64.57539
4	INT	64.16718
5	INT	66.30892

```r
res1 <- TOSTER::t_TOST(
  formula = weight ~ group,
  data = weight_data,
  eqb = 2,  # equivalence bounds of ± 2 kg
  smd_ci = "t")  # t-distribution
```

Result 1

	t	SE	df	p.value
t-test	1.833429	0.4129882	197.8243	6.824144e-02
TOST Lower	6.676183	0.4129882	197.8243	1.208527e-10
TOST Upper	-3.009324	0.4129882	197.8243	1.479486e-03



The equivalence test was significant, t(197.82) = −3, p < 0.01
The null hypothesis test was non−significant, t(197.82) = 1.833, p = 0.07

```r
TOSTER::describe(res1)
# "Using the Welch Two Sample t-test,
# a null hypothesis significance test (NHST),
# and a equivalence test, via two one-sided
# tests (TOST), were performed with an
# alpha-level of 0.05. These tested the null hypotheses tha
# to 0 (NHST), and true mean difference is more
# extreme than -2 and 2 (TOST). The equivalence
# test was significant, t(197.824) = -3.01,
# p = 0.001 (mean difference = 0.757 90%
# C.I.[0.075, 1.44]; Hedges's g(av) = 0.258
# 90% C.I.[0.022, 0.494]). At the desired
# error rate, it can be stated that the
# true mean difference is between -2 and 2."
```

Example 2

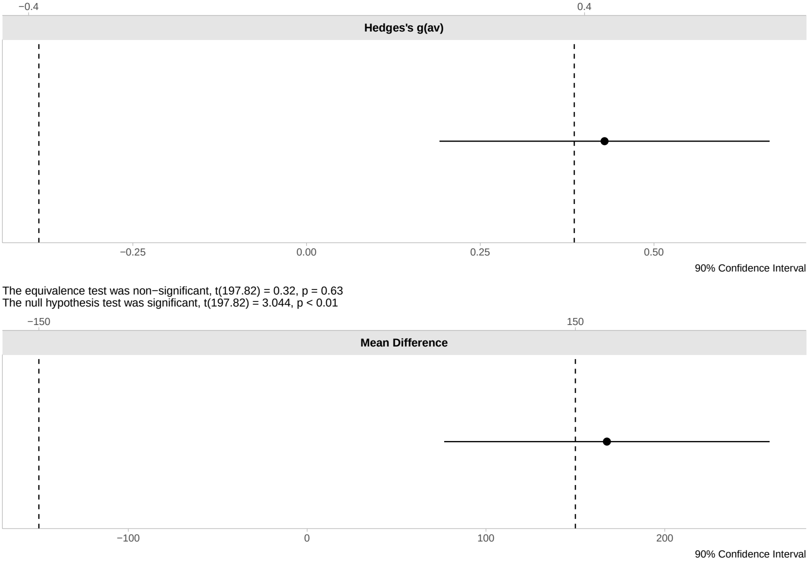
- Suppose we want to test whether two interventions are equally effective in improving birth weight among undernourished pregnant women
- The equivalence bound ($\triangle$) is set to $\pm$ 150 gr

```r
set.seed(0009) # for reproducibility

# create two groups
group <- rep(c("INT", "CON"), each = 100)

# Simulate data
birth_weight_data <- data.frame(
  group,
  weight = ifelse(
    group == "INT",
    rnorm(n = 100, mean = 3200, sd = 400),
    rnorm(n = 100, mean = 3400, sd = 400)
    )
  )
```

```r
res2 = TOSTER::t_TOST(
  formula = weight ~ group,
  data = birth_weight_data,
  eqb = 150,  # equivalence bounds of ± 150 kg
  smd_ci = "t")  # t-distribution
```

Result 2

	t	SE	df	p.value
t-test	3.0441177	55.06509	197.8243	2.650925e-03
TOST Lower	5.7681665	55.06509	197.8243	1.524578e-08
TOST Upper	0.3200688	55.06509	197.8243	6.253730e-01


−0.4
0.4
Hedges's g(av)
−0.25
0.00
0.25
0.50
90% Confidence Interval
The equivalence test was non−significant, t(197.82) = 0.32, p = 0.63
The null hypothesis test was significant, t(197.82) = 3.044, p < 0.01
−150
150
Mean Difference
−100
0
100
200
90% Confidence Interval

```r
TOSTER::describe(res2)
# "Using the Welch Two Sample t-test,
# a null hypothesis significance test
# (NHST), and a equivalence test,
# via two one-sided tests (TOST),
# were performed with an alpha-level
# of 0.05. These tested the null
# hypotheses that true mean difference
# is equal to 0 (NHST), and true mean
# difference is more extreme than -150
# and 150 (TOST). The equivalence test
# was not significant (p = 0.625). The
# NHST was significant, t(197.824) = 3.044,
# p = 0.003 (mean difference = 167.625
# 90% C.I.[76.624, 258.625];
# Hedges's g(av) = 0.429 90%
# C.I.[0.191, 0.666]). At the desired error
# rate, it can be stated that the true mean
# difference is not equal to 0
# (i.e., no equivalence)."
```

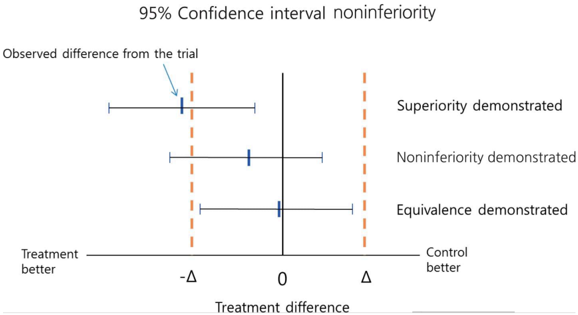
- $\triangle_L$ and $\triangle_U$ should be specified a priori, before collecting data
- This is essential because wider bounds make it easier to conclude equivalence
- The statistical power of an equivalence test depends on the observed difference and the equivalence bounds: smaller bounds and smaller effects require larger sample sizes
- Check this link out for illustrations

Non-inferiority test

- A non-inferiority test is performed to test whether a new intervention is not significantly worse than a control by more than a pre-defined, acceptable margin ($\triangle$)
- Rather than aiming to show superiority or equivalence, it assesses whether the new intervention is "good enough"
- In clinical settings, a non-inferiority test is used when the new treatment may offer important advantages over currently available standard treatments, in terms of improved safety or cost.

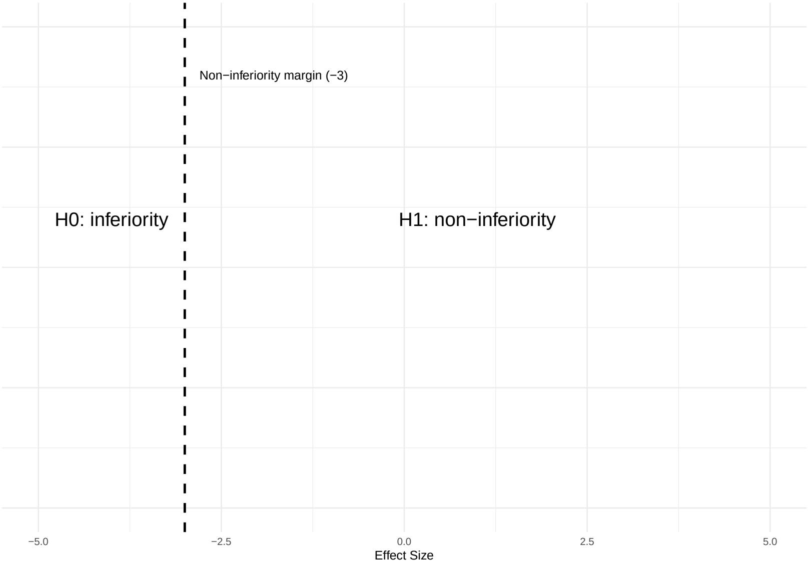
H_0 in a non-inferior test

- When testing for equivalence, we test whether a treatment effect is inside a prespecified equivalence margin [-$\triangle$, $\triangle$].
- Similarly, when testing if a treatment is at least not worse than another treatment, we test if the effect is above a prespecified non-inferiority margin -$\triangle$.
- H_0: $\mu_C - \mu_T \leq \triangle$ (Treatment (T) is inferior to the Control (C) by $\triangle$ or more)
- H_1: $\mu_C - \mu_T > \triangle$ (Treatment (T) is inferior to the Control (C) by less than $\triangle$)
- The goal is to reject values smaller than below -$\triangle$


95% Confidence interval noninferiority
Observed difference from the trial
Superiority demonstrated
Noninferiority demonstrated
Equivalence demonstrated
Treatment better
Control better
-Δ
0
Δ
Treatment difference

Example non-inferior test

- Suppose we want to test whether a new intervention is not worse than a standard intervention in reducing depression, as measured on a scale from 1 to 20
- Higher scores indicate greater severity
- The non-inferiority margin ($\triangle$) is set to - 3, meaning the new intervention (I) can be at most 3 points worse than the standard intervention (C) and still be considered non-inferior
- H_0: μ_C - $\mu_I \leq$ - 3
- H_1: μ_C - $\mu_I >$ - 3


Non−inferiority margin (−3)
H0: inferiority
H1: non−inferiority
−5.0
−2.5
0.0
2.5
5.0
Effect Size

```r
set.seed(0009) # for reproducibility

# create two groups
group <- rep(c("INT", "CON"), each = 100)

# Simulate data
depression_data <- data.frame(
  group,
  depression = ifelse(
    group == "INT",
    rnorm(n = 100, mean = 14, sd = 2),
    rnorm(n = 100, mean = 12, sd = 2)
    )
  )
```

95% CI approach

- A two-sided 95% CI can be used to assess non-inferiority
- When $\alpha = 0.05$, then corresponding two-sided CI is 1–$\alpha =$ 95%
- To evaluate non-inferiority, compare the lower bound of the 95% CI with the non-inferiority margin
- The lower bound of a 95% CI is equivalent to the bound from one-sided test at $\alpha = 2.5$ (a one-sided 97.5% CI)
- Type I error is 2.5%
- If the lower bound of the 95% CI lies above the non-inferiority margin, non-inferiority is demonstrated

```r
res3 <- t.test(depression ~ group, depression_data)

broom::tidy(res3) |>
  dplyr::select(estimate, conf.low, conf.high) |>
  kableExtra::kable()
```

estimate	conf.low	conf.high
-2.161877	-2.704826	-1.618927

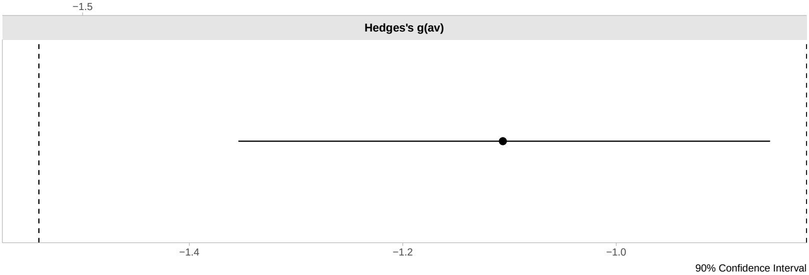
The lower bound of the 95% CI is above -3 ($\triangle$) so non-inferiority is demonstrated

Non-inferiority test with TOST (90% CI) approach

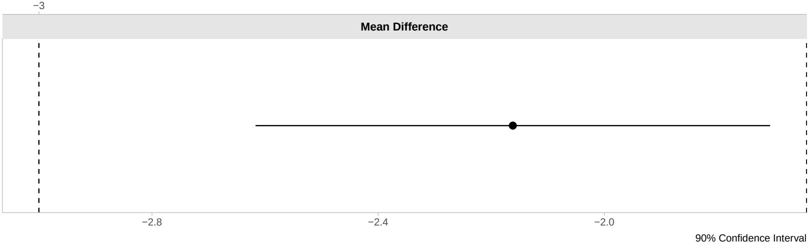
- TOST procedure involves conducting two one-sided t-test, each at $\alpha = 0.05$
- We calculate a 90% CI (1 - 2α)
- For non-inferiority, we take the result of the one-sided t-test that evaluates whether the effect is greater than the non-inferiority margin
- Type I error is 5%
- If the lower bound of the 90% CI lies above the non-inferiority margin, non-inferiority is demonstrated.

```r
res3 <- TOSTER::t_TOST(depression ~ group,
                       depression_data,
                       low_eqbound = -3,
                       high_eqbound = Inf,
                       paired = FALSE,
                       alpha = 0.05)
```

	t	SE	df	p.value
t-test	-7.852078	0.2753255	197.8243	0.0000000
TOST Lower	3.044118	0.2753255	197.8243	0.0013255
TOST Upper	-Inf	0.2753255	197.8243	0.0000000



The equivalence test was significant, $t(197.82) = 3.04$, $p < 0.01$
The null hypothesis test was significant, $t(197.82) = -7.85$, $p < 0.01$



References

Lakens, Daniël. 2017. "Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses." *Social Psychological and Personality Science* 8 (4): 355–62. https://doi.org/10.1177/1948550617697177.

Schuirmann, D. J. 1987. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability." *Journal of Pharmacokinetics and Biopharmaceutics* 15 (6): 657–80. https://doi.org/10.1007/BF01068419.

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